

Experiment No 06 :- Potentiometric Titration.

Theory :-

- Redox reaction

- Graphs

• E_{cell} vs V

• $\Delta E/\Delta V$ vs $V + \Delta V/2$

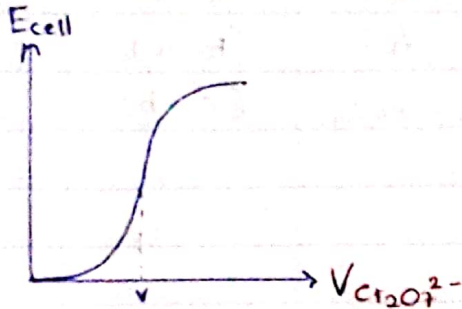


Figure No 01 :- Title

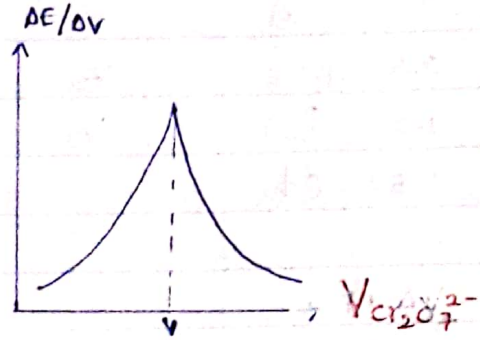


Figure No 02 :- Title

Procedure :-

(Use past participle)

(draw Setup of the experiment)

Figure No 03 :- Setup.

Observations & Results :-

Table No 01 :- The cell EMF measurements for each $\text{K}_2\text{Cr}_2\text{O}_7$ volumes

$V_{\text{K}_2\text{Cr}_2\text{O}_7}$ (± 0.05) cm^3	cell EMF ($\pm 0.001 \text{ V}$)
0.00	
0.50	
1.00	

0.00 $\rightarrow$ 25.00

25.00

Calculations :-

Table No 02 :- Calculations for $\Delta E/\Delta V$ for potential changes at each $\text{Cr}_2\text{O}_7^{2-}$ volume added.

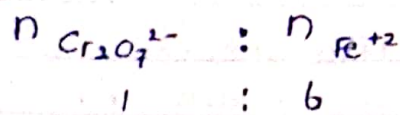
$V_{\text{K}_2\text{Cr}_2\text{O}_7}$ (± 0.05) cm^3	EMF (± 0.001) V	ΔE	ΔV	$\Delta E/\Delta V$
0.00 (a)	V_1	$(V_1 - V_1)$	$(a - a)$	
0.50 (b)	V_2	$(V_2 - V_1)$	$(b - a)$	
1.00 (c)	V_3	$(V_3 - V_2)$	$(c - b)$	
1.50 (d)				
2.00				
2.50				
3.00				
3.50				
4.00				
4.50				
5.00				
5.50				
6.00				
6.50				
7.00				
7.50				
8.00				
8.50				
9.00				
9.50				
10.00				

Graph (2)

Oxidation reaction.

Reduction reaction.

Overall reaction.



The equivalence point of the titration (V_0) = $x \text{ cm}^3$

$$C_{\text{K}_2\text{Cr}_2\text{O}_7^{2-}} = 0.01 \text{ M}$$

$$n_{\text{K}_2\text{Cr}_2\text{O}_7} = CV \\ = \underbrace{0.01 \text{ mol dm}^{-3}}_{a'} \times x \times 10^{-3} \text{ dm}^3$$

$$n_{\text{Fe}^{+2}} = (a' \times 6) \text{ mol}$$

$$C_{\text{Fe}^{+2}} = \frac{a' \times 6}{20 \times 10^{-3}} \\ = \text{mol dm}^{-3}$$

conclusion:-

concentration of Fe^{+2} solution is ----- //

Discussion:-

- Potentiometric titration.
- Types of potentiometric titration.
- Short note of
 - Ag/AgCl calomel electrode
 - pt electrode
- Why use H_2SO_4 in this experiment
- What is the indicator that use for this experiment
- Errors of potentiometric titration.
- Applications of potentiometric titrations

References :-

— (3)